

ASSESSMENT INSTRUCTIONS

COURSE INFORMATION

Course Code	CDE2313	Programme	Bachelor in Data Science
Course Title	Mobile Application Development	Academic Session	2025/2026
Classification	Field Electives	Semester	3
Lecturer	Ts Mohd Zulkifli Mohd Zaki	Venue	B09-301, B09-210

ASSESSMENT 2

Method	Group Project (2-3 members)	Weight (%)	25
Course Learning Outcome/s	CLO3: Build a simple mobile application using Android Studio. (P5, PLO3)		
Core Value/s	-	3-Zeroes	-

INSTRUCTIONS

Purpose of Assessment:

The purpose of this assessment is to evaluate students' ability to design, develop, test, and demonstrate a functional Android mobile application using Android Studio to solve a real-world problem. The assessment emphasizes authentic software engineering practices and encourages students to integrate technical knowledge, problem-solving skills, teamwork, and responsible use of Artificial Intelligence (AI) throughout the software development lifecycle.

Students are expected to demonstrate competence in developing a complete Android application with minimal supervision, reflecting the requirements of Psychomotor Level P5 (Naturalization). At this level, students should be able to independently perform software development tasks, apply appropriate programming techniques, debug and refine their applications, and justify their technical decisions during the project demonstration and technical viva. This assessment aligns with the following Course Learning Outcome, which students should be able to build a simple mobile application using Android Studio. The assessment also supports Program Learning Outcome by enabling students to demonstrate practical software development skills in an authentic learning environment.

Task Description:

Working in **groups of two (2) to three (3) students**, each group is required to **design, develop, test, and demonstrate a functional Android mobile application using Android Studio** based on a provided case study.

All groups will work from the provided case study, which includes background problems, user requirements, and expected system functionalities. Students are expected to analyze the case study, interpret the requirements, and implement an Android application that addresses the identified needs.

The project should follow an iterative software development process, including requirements analysis, interface design, application development, testing, debugging, refinement, and presentation. Students are encouraged to demonstrate creativity through interface design, user experience, feature enhancements, and implementation approaches, provided the core requirements are fulfilled.

The project aims to evaluate students' ability to apply Android development concepts, software engineering practices, and collaborative problem-solving skills to produce a functional mobile application.

- **Scope:**

The project aims to evaluate students' ability to apply Android development concepts, software engineering practices, and collaborative problem-solving skills to produce a functional mobile application.

The project scope is determined by the **case study supplied by the lecturer (Please refer to the case study description attached separately)**

Each case study will include:

- Background of the organization or problem domain.
- Description of the current challenges.
- Target users.
- Functional requirements.
- Non-functional requirements.
- Business rules.
- Expected application features.
- Optional enhancement requirements.

Students are required to analyze the case study and implement an Android application that satisfies the specified requirements. The completed application should demonstrate appropriate use of Android Studio and relevant Android development components while maintaining good software engineering practices.

- **Structure/Format:**

The GitHub repository should follow the structure below.

```
MobileAppProject/
├── README.md                # Project landing page
├── LICENCE
├── .gitignore
├── app/                     # Android Studio Project
├── gradle/                 # Gradle configuration
├── screenshots/            # Screenshots for the mobile app
├── docs/                   # Related documentations
│   ├── Case_Study_Analysis.md
│   ├── wireframe/
│   └── diagrams/
├── apk/                    # Release-ready APK file
│   └── app-release.apk
├── presentation/           # Presentation slides for final demonstration
│   ├── Presentation.pptx
│   └── Presentation.pdf
├── ai/                     # Concise AI usage declaration
│   └── AI_Usage_Log.pdf
└── assets/                 # Store supporting resources
```

The repository should demonstrate the complete software development process based on the assigned case study. The repository must contain a professional README file including:

- Project title
- Course information
- Group Members (Student Name, Student ID)
- Case Study – Problem overview, summary of implemented solutions
- Key implemented Features
- Technology Stack and Application Architecture
- Installation Guide
- How to Build the Project
- APK Download
- Screenshots
- GitHub Repository Structure
- Acknowledgements
- References (if applicable)

The README should serve as the project's landing page.

- **Specific Requirements:**

Students are required to implement all mandatory functional requirements specified in the case study. The application must also demonstrate competency in Android Studio by incorporating appropriate Android components.

Each group shall develop a functional Android mobile application using Android Studio based on the provided case study. The application shall demonstrate the practical application of Android development concepts, software engineering principles, and collaborative development practices. The project shall satisfy the following requirements.

- **Functional requirements** – The developed application must implement all mandatory functional requirements specified in the assigned case study.
- **Technical Requirements** – The application must be developed using Android Studio and demonstrate competency in Android application development.
- **Software Engineering Requirements** – Students are expected to demonstrate good software engineering practices throughout the project.
- **GitHub Repository Requirements** – The GitHub repository serves as the primary submission and evidence of project development. The repository shall contain a comprehensive README.
- **Git Commit Requirements** – To demonstrate continuous software development, each group is expected to maintain a consistent Git commit history.
- **User Interface Requirements** – The application should provide a consistent and user-friendly interface.
- **Testing Requirements** – Students shall perform functional testing before submission.
- **Presentation and Demonstration Requirements** – Each group shall deliver a 10–15-minute presentation and demonstration. Every student will participate in an individual technical viva.

- **Group/Individual Work:**

This is a **group-based assessment**.

Each group is expected to:

- Analyze the provided case study to understand the problem domain and user requirements.
- Identify the required system functionalities and constraints described in the case study.
- Design appropriate mobile application architecture and user interface.
- Develop the application using Android Studio.
- Implement all mandatory functional requirements.
- Apply appropriate Android development components and programming techniques.
- Test and debug the application to ensure reliability and usability.
- Maintain a GitHub repository throughout the project, demonstrating continuous development through meaningful commits.
- Present and demonstrate the completed application during the scheduled assessment session.
- Individually explain their technical contributions and implementation decisions during the technical viva.

Each student must demonstrate individual contribution through:

- Git commit history
- Code ownership
- Pull requests (if used)
- Issues completed
- Participation during demonstration
- Individual technical viva
- Marks may be adjusted based on individual contribution if there is significant imbalance within the group.

Each group must submit the following through AIU Moodle LMS:

- **GitHub Repository URL**
- **PDF document containing GitHub Repository URL, Android Developer Profiles URL for each member, and Badge Summary**
- **Repository must remain accessible until final grades are released.**

File Format	URL, PDF	Submission Platform	AIU Moodle LMS
Word Count/Length	-	Submission Deadline	Week 6 (18 September 2026, by or before 11:59 PM)
Citation Style	APA 7th Edition	Late Submission Policy	5 points deduction per day

ACADEMIC INTEGRITY

Plagiarism:	This assignment should be checked using Turnitin for originality. The acceptable level of collaboration should not exceed 30%.
Generative AI:	In this assessment, you must not use generative artificial intelligence (AI) to generate any materials or content in relation to the assessment task.

The use of Generative Artificial Intelligence (GenAI) tools (e.g., ChatGPT, GitHub Copilot, Gemini, Claude) is permitted as a learning support tool, but not as a substitute for students' own analytical thinking or practical work.

Students may use AI responsibly:

- Learning Android programming concepts
- Understanding Android APIs
- Debugging code
- Explaining programming concepts
- Generating small code examples
- Improving code readability
- Refactoring existing code
- Generating UI design ideas

Students must not:

- Generate and submit an entire application using AI without understanding the implementation.
- Present AI-generated work as entirely their own.
- Use AI to answer questions during the live demonstration or individual technical viva unless explicitly permitted.
- Fabricate Git commit histories or repository activity.

Each group shall submit an **AI Usage Log** documenting:

- AI tool(s) used
- Purpose of use
- Representative prompts (where appropriate)
- AI-generated suggestions adopted or rejected
- Modifications made by the students
- Reflection on how AI supported their learning and development process

Students remain fully responsible for the correctness, quality, and integrity of all submitted work, including any code or content generated with AI assistance.

All AI-assisted work must be reviewed, modified, tested, and understood by the students. Students should be prepared to explain any part of their repository if requested by the lecturer. Failure to demonstrate ownership of the submitted work may result in penalties under the University's Academic Integrity Policy.

RUBRIC

Criteria	Excellent (5)	Very Good (4)	Good (3)	Satisfactory (2)	Poor (1)	Not Demonstrated (0)
Problem Analysis & Case Study Implementation (Weight: 10%)	Demonstrates an excellent understanding of the assigned case study. All mandatory and relevant optional requirements are accurately interpreted and fully implemented. The solution effectively addresses the identified problem with clear evidence of thoughtful analysis.	Demonstrates a strong understanding of the case study. All mandatory requirements are implemented correctly with only minor omissions or interpretation issues.	Demonstrates an adequate understanding of the case study. Most mandatory requirements are implemented, although some functionality or interpretation is incomplete.	Demonstrates a basic understanding of the case study. Several mandatory requirements are partially implemented or inadequately addressed.	Demonstrates limited understanding of the case study. Only a few requirements are implemented correctly.	No meaningful problem analysis or solution design submitted.
Mobile Application Functionality (Weight: 25%)	All application features function correctly without errors. Navigation, data processing, validation, CRUD operations, and user interactions operate smoothly. The application is stable, complete, and ready for deployment.	Most application features function correctly with only minor issues that do not significantly affect usability.	Core application features function adequately, although some features contain minor bugs or incomplete implementation.	Basic application functionality is available, but several important features are incomplete or unreliable.	Application has limited functionality and major defects that significantly affect usability.	Application cannot be executed or contain little or no functional implementation.

Criteria	Excellent (5)	Very Good (4)	Good (3)	Satisfactory (2)	Poor (1)	Not Demonstrated (0)
Android Studio Implementation & Code Quality (Weight: 20%)	Excellent use of Android Studio and Android components. Code is modular, well organized, readable, maintainable, and follows good programming practices. Appropriate use of architecture, resource management, and error handling is evident.	Good implementation using appropriate Android components. Code is generally well structured with only minor issues related to organization or maintainability.	Appropriate implementation with acceptable coding practices. Some inconsistencies or code duplication exist but do not severely affect maintainability.	Basic implementation with inconsistent coding style, limited use of Android components, or weak project organization.	Poor implementation with disorganized code, inappropriate use of Android components, or numerous coding issues.	Source code is missing, incomplete, or demonstrates no meaningful implementation.
GitHub Repository Management & Version Control (Weight: 10%)	Repository is exceptionally organized with a comprehensive README, logical folder structure, regular meaningful commits from all members, clear version history, and professional repository management practices.	Repository is well organized with a complete README and consistent commit history demonstrating active collaboration.	Repository contains all required files with an acceptable README and adequate commit history. Most members contribute regularly.	Repository is functional but lacks organization, complete documentation, or consistent development history.	Repository is poorly organized with minimal commits or evidence of collaboration. README is incomplete or unclear.	Repository is missing, inaccessible, or contains no meaningful development evidence.
User Interface (UI) & User Experience (UX) Design (Weight: 10%)	User interface is professional, visually appealing, intuitive, responsive, and follows the principles. Navigation is seamless and enhances the user's experience.	The user interface is attractive and user-friendly, with only minor usability issues.	User interface is functional and generally easy to use, although some inconsistencies exist.	User interface provides only basic usability with noticeable layout or navigation issues.	User interface is confusing, inconsistent, or difficult to use.	User interface is absent or unusable.

Criteria	Excellent (5)	Very Good (4)	Good (3)	Satisfactory (2)	Poor (1)	Not Demonstrated (0)
Testing & Debugging (Weight: 5%)	Application has been thoroughly tested with comprehensive validation of features. Errors are effectively handled, and debugging evidence demonstrates a stable application.	Application has been well tested with only minor unresolved issues.	Application has been adequately tested, covering most major functions.	Limited testing has been conducted, leaving several unresolved issues.	Minimal testing is evident with many unresolved defects.	No evidence of testing or debugging.
Live Demonstration (Weight: 10%)	Presentation is highly professional, well organized, and delivered confidently. The application is demonstrated smoothly, and all features are clearly explained with effective responses to questions.	Clear and well-structured presentation with a successful demonstration of most application features. Questions are answered accurately.	Satisfactory presentation demonstrating the main application features with acceptable explanations.	Presentation lacks clarity or organization, and several features are not demonstrated effectively.	Presentation is incomplete, poorly delivered, or demonstrates limited understanding of the application.	No presentation or live demonstration conducted.
Individual Technical Viva (Weight: 5%)	Student demonstrates comprehensive understanding of their contribution, implementation decisions, Android components, programming logic, debugging process, and software architecture. All questions are answered accurately and confidently.	Students demonstrate strong understanding of their contribution with only minor inaccuracies.	Students demonstrate satisfactory understanding of their contribution and implementation decisions.	Students demonstrate limited understanding and require significant prompting to explain their work.	Students are unable to explain most aspects of their contribution or technical implementation.	Students do not attend the viva or are unable to demonstrate any understanding of the submitted work.

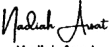
Criteria	Excellent (5)	Very Good (4)	Good (3)	Satisfactory (2)	Poor (1)	Not Demonstrated (0)
Responsible AI Usage & Professional Practice (Weight: 5%)	AI is used responsibly and transparently to support learning and development. AI Usage Log is complete, reflective, and demonstrates critical evaluation of AI-generated suggestions. Professional collaboration and ethical conduct are consistently demonstrated.	Responsible AI usage is evident with a complete AI Usage Log and good professional practices.	Appropriate AI usage is documented with adequate reflection and acceptable professional conduct.	AI usage is partially documented with limited reflection or inconsistent professional practices.	AI usage is poorly documented or raises concerns regarding responsible use or professional conduct.	AI Usage Log is missing, or there is evidence of academic misconduct or inappropriate AI usage.
Total Points:			100			

APPROVAL LEVEL

Approved by:

Nadiah Arsat

Head of Programme


Nadiah Arsat
Deputy Dean, School of Computing & Informatics
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Signature

3/8/26

Date